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Reciprocal replacement and the maintenance of codominance in a beech-maple forest

Kerry D. Woods

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Regeneration of dominant species in an undisturbed beech-maple forest in Michigan, USA, was studied. Spatial distributions of saplings of beech and maple were related to influence by canopy trees of these species. Some factor effecting a differential mortality of young individuals of the two species was indicated and it is suggested that this factor is related to species of nearby canopy trees. This cause of mortality appeared to act on individuals between one and four meters in height. Assuming that replacement of fallen trees in such a forest is primarily by advance reproduction in the form of such seedlings and saplings as studied here, it appears that individuals of either dominant species will tend to be replaced by individuals of the other species in a pattern of reciprocal replacement. This, then, is a possible mechanism for indefinite maintenance of codominance in a climax forest.

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Исследовали восстановление доминирующих видов в ненарушенном букво-кленовом лесу в Мичигане (США). Пространственное распределение всходов бука и клена зависит от густоты крон этих пород. Установлены факторы дифференциальной гибели проростков обоих видов: в основном это видовой состав соседствующих деревьев. Эта причина влияет на выживаемость растений высотой 1–4 м. Принимая во внимание, что замена погибших деревьев в таком лесу определяется скоростью самосева и подроста, имеется тенденция к замене доминирующих видов другими видами в зависимости от характера распределения. Это, т.о., возможный механизм неизбирательного поддержания содоминантов в климаксовом лесу.

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1. Introduction

The coexistence of two dominant climax, or tolerant, canopy species in a forest stand with relatively uniform substrate might, at first consideration, seem at variance with the requirements of the competitive exclusion principle. Such coexistence presumably implies some significant difference between the niches of the codominant species – that is, some difference in the factors limiting the two species (Levin 1970). Workers have often been frustrated in finding such differences through studies of mature individuals of two such species. In such cases it might be useful to study the often neglected niche variables of early developmental stages. An environment perceived as being nearly uniform by mature trees may appear to a young individual – a seed, seedling, or sapling – to be highly variable. The great importance of these early stages of the life cycle in defining a species' niche has been taken into account in the various concepts of regeneration niche (Grubb 1977), safe-site (Harper 1977), and microsite mosaic (Whittaker and Levin 1977).

In order to differentiate niches of several species in this way one must find some environmental factor causing variability which is experienced by young individuals but not by mature ones. Many microsite factors might fit this requirement if they vary on a spatial scale small enough that they may be "averaged" by a mature individual. One such factor whose influence is uniquely felt by small individuals is canopy influence. Influence by canopy individuals is, then, a potentially important niche variable for young trees.

Jones (1945) has suggested that, in forests at demographic equilibrium and dominated by shade-tolerant species, disturbance is normally on the scale of a single tree or a small number of trees and replacement is usually by advance reproduction in the form of suppressed saplings of the dominant species. For such forests, where codominance is by two species, a mechanism might be proposed whereby indefinite coexistence of these two species might be effected. If one or both species show, as seedlings or saplings, superior survival in sites dominated by canopy trees of the other species, the two may tend to replace each other, thus maintaining codominance through a pattern of what I shall term *reciprocal replacement*. This sort of dynamic interaction has been hinted at previously as early as Aubreville who observed that in certain forests, "any one tree species never regenerated under itself," (Grubb 1977). Jones (1945) cites, "a very strong belief amongst continental foresters in the tendency of one species to regenerate beneath canopy of the other". Schaeffer and Moreau (1958) reported extensively on cases of this type of phenomenon. Whittaker and Levin (1977) discuss the role of the "microsite mosaic" in such equilibrium mechanisms. Watt (1947) suggests a similar process producing his "phasic equilibrium."

This study addresses itself to the possible occurrence

of reciprocal replacement in a stand of beech-maple (*Fagus grandifolia* Ehrh. – *Acer saccharum* Marsh.) forest in Michigan, USA. In order to test for this possibility the assumption is made that a canopy tree will most probably be replaced by a sapling growing beneath it. Given this assumption, data on the relative distribution of beech and maple canopy trees and saplings may be used to test the null hypothesis H_0 : Beech and maple saplings are similarly distributed with respect to canopy species influence.

Rejection of this hypothesis will indicate that saplings of one or both species are established or survive preferentially beneath canopy influence of a particular species. If favourable canopy influence is of the same species as that of the sapling same-species replacement is suggested; if of the other species, reciprocal replacement is indicated.

H_0 will be rejected, with reciprocal replacement being the alternate hypothesis suggested, and mechanisms possibly causing the phenomenon will be discussed.

2. Materials and methods

2.1. Study Site

Samples for this study were taken in Warren Woods, a part of Warren Dunes State Park in Berrien Co., Michigan. This stand is an essentially undisturbed remnant of beech-maple forest. Its vegetation, history, and soil characteristics have been described by Cain (1935). Cain's compositional description and that obtained from this study are quite similar, suggesting that the forest is indeed near demographic equilibrium. It is also near the composition described by Braun (1950) as typical of the beech-maple forest type. The topography of Warren Woods conforms to that described by Braun for the till plain and lake district of southern Michigan as having, "low relief where minor differences in elevation . . . determine the forest type." Areas of relatively greater elevation and better drainage are strongly dominated by beech and maple. Samples were taken in such areas.

The proportions of basal area represented by beech (65%) and maple (35%) in this study and the negligible contribution of other species are typical of the mature beech-maple forest as described by Braun. Scattered individuals of *Ostrya virginiana* (Mill.) K. Koch, *Acer rubrum* L., and *Liriodendron tulipifera* L. occurred in the canopy size class. The rarity of these secondary species supports the hypothesis that canopy replacement is mostly by saplings of the two dominant species. Brewer and Merritt (1978) have studied wind throw and replacement in Warren Woods and suggest that, in fact, in small gaps replacement is almost exclusively by beech and maple saplings.

Understory species with frequencies of more than 10% in understory sub-plots were *Arisaema triphyllum* (L.) Schott, *Asimina triloba* (L.) Dunal, *Carex plantaginea* Lam., *Dentaria laciniata* Muhl., *Epifagus vir-*

giniana (L.) Bart., *Euonymus obovatus* Nutt., *Galium aparine* L., *Osmorhiza claytoni* (Michx.) C. B. Clarke, *Parthenocissus quinquefolia* (L.) Planch., *Polygonatum pubescens* (Willd.) Pursh, *Sambucus pubens* Michx., *Sanicula marilandica* L., *Viola pubescens* Ait., and *Viola* sp. Other species were *Carex platyphylla* Carey, *Caulophyllum thalictroides* (L.) Michx., *Dryopteris spinulosa* (L.) F. Muell.) Watt., *Lindera benzoin* (L.) Blume, *Mitchella repens* L., *Rhus radicans* L., *Ribes cynosbati* L., *Sambucus canadensis* L., *Sanguinaria canadensis* L., *Smilax tamnoides* Var. *hispida* (Muhl.) Fern., and *Viburnum acerifolium* L.

2.2. Sampling

Data were gathered from six 20 m × 50 m (0.1 ha) quadrats in Warren Woods in June, 1976. Quadrats were placed so as to avoid inclusion of extensive disturbance (e.g. unusually large tree-fall gaps, paths, and vehicle trails) or pronounced discontinuity or variation of habitat and vegetation type (e.g., unstable banks, flood plains, and marshy areas). In each quadrat species, diameter at breast height (dbh), distance to nearest neighbour, and dbh and species of nearest neighbour were recorded for each canopy tree. Canopy individuals were arbitrarily defined as those with dbh greater than 12.5 cm.

Species, basal diameter, height, average radius of crown, substrate condition, and apparent condition of health were recorded for each sapling (individuals of greater than two meters height and smaller in dbh than canopy category) within the quadrats. When possible it was determined, in the case of beech, whether the sapling was derived from a root sprout or seedling. Distance and direction to canopy trees nearer than 10 m and their species and dbh were recorded for each sapling. Presence and cover estimates for understory species and age (determined by counting terminal-bud scars) of seedlings of beech and maple were recorded for sapling-centered circles of 1 m Ø for each sapling. In quadrats four, five, and six, a subjective judgment was made of predominant canopy species at the site of each dead sapling of beech or maple.

In each 0.1 ha quadrat, fifty contiguous 1 m² plots were located along the long axis of the quadrat. Presence and cover of understory species and substrate conditions were recorded and beech and maple seedlings were aged for each plot.

2.3. Analysis

In order to test H₀ several indices were devised to express, for each sapling, some quantitative measure of influence of canopy trees of the two dominant species. It is desired that such indices should take into account (1) all canopy individuals within a set distance from the sapling, (2) size of these various canopy trees, and (3) distance to these trees from the sapling.

The first such index used in this study was that proportion of total beech and maple basal area within 10 m of the sapling represented by beech:

$$I_1 = \frac{\Sigma \text{Beech basal area within 10 m of sapling}}{\Sigma \text{Beech and maple basal area within 10 m of sapling}}$$

This index accounts for all neighbourhood trees and their size, but does not permit consideration of difference in absolute magnitudes of canopy influence nor is it distance-weighted. A second index, intended to satisfy these shortcomings, but somewhat more contrived, was developed:

$$I_2 = \frac{\sum_{\text{distance}}^b \text{dbh}}{\sum_{\text{distance}}^m \text{dbh}} - \frac{\sum_{\text{distance}}^m \text{dbh}}{\sum_{\text{distance}}^b \text{dbh}}$$

where b is the number of beech canopy trees within 10 m of a sapling, m the number of maple canopy trees, and distance that from the sapling to each canopy tree. Thus, given the assumption that these quotients adequately describe a tree's influence on a microsite, I₂ will take a value near zero when the two canopy species are nearly equal in influence on a given sapling, become positive with increasing beech influence, and negative with increasing maple influence.

Values of these indices were calculated for each sapling and subjected to statistical treatment. In order to test the null hypothesis all values were ordered for each index and rank-sum tests were performed on these rankings separating saplings into two categories by species. Significant results of such tests should then indicate differential response of the two species as saplings to whatever aspect of canopy influence is represented by the index.

All understory species with more than ten occurrences in the sapling-centered plots were tested, using chi-squared tests, for departure from random assortment between those circles centered on beech and those centered on maple saplings.

Seedlings of the two dominant species occurring in sapling-centered plots were similarly tested, with quadrats divided into several abundance classes for maple seedlings and into presence and absence classes for beech seedlings. Rank-sum tests were also performed by dividing quadrats into two groups by abundance of maple seedlings and again by presence or absence of beech seedlings and ranking by the value of one the previously described indices for the sapling on which the plot was centered. This tests the null hypothesis that seedlings of beech and maple are similarly distributed with respect to species of canopy influence, though there is a possibility here of interaction with sapling dynamics.

Single-species clumping of canopy individuals was tested for using the segregation index derived by

Tab. 1. Basal area for canopy trees and saplings in m². ha⁻¹ by species and quadrat.

Species	Quadrat						Average
	1	2	3	4	5	6	
<i>Fagus</i>	23.7	35.7	28.6	27.8	29.1	10.0	25.8
<i>Acer</i>	17.5	13.6	15.5	9.8	2.4	23.1	13.6
Other	0.0	0.3	0.2	0.1	0.0	0.5	0.2
Total	41.2	49.6	44.3	37.7	31.5	33.6	39.6

Tab. 2. Sapling size distribution by species and quadrat. “Other” species include *Hamamelis virginiana*, *Ostrya virginiana*, *Fraxinus americana*, *Ulmus rubra*, *Quercus borealis*, *Liriodendron tulipifera*, and *Acer rubrum*.

Quadrat	Species	Diam. class (cm)				
		0–2.5	2.5–5.0	5.0–7.5	7.5–10.0	10.0–12.5
1	<i>Acer</i>	8	13	11	12	4
	<i>Fagus</i>		4	5	6	3
	Other		1	1		
2	<i>Acer</i>	1	2	9	13	10
	<i>Fagus</i>	6	5	3	4	6
	Other			2	3	
3	<i>Acer</i>	5	13	14	15	5
	<i>Fagus</i>	2	8	11	1	5
	Other	2	1		3	1
4	<i>Acer</i>			3	7	6
	<i>Fagus</i>	3	9	8	5	6
	Other					1
5	<i>Acer</i>	3	3	1	5	2
	<i>Fagus</i>		1	1	3	4
	Other				1	
6	<i>Acer</i>		4	6	13	7
	<i>Fagus</i>	2	9	15	3	4
	Other			1		2
Total	<i>Acer</i>	17	35	44	65	34
	<i>Fagus</i>	13	36	43	22	28
	Other	2	2	4	7	4
Total	All spp.	32	73	91	94	66
	Average ha ⁻¹	53	122	152	154	110

Peterson (1976). The index is the ratio of the probability of finding the nearest neighbour of an individual to be its conspecific to the probability of a second tree randomly chosen from the stand being of the same species. These probabilities were estimated by ratios of conspecific nearest neighbour pairs to total numbers of pairs and species ratios in the samples taken.

Statistical analyses were performed as described in Snedecor and Cochran (1967). Computer programs for their performance and other compilations of data were written by the author.

3. Results

General vegetational data from Warren Woods are summarized in Tabs 1, 2 and 3. The estimate of 29.6 m² ha⁻¹ for canopy basal area is a high figure for north-eastern deciduous forests, indicating a highly developed, mature forest (Whittaker 1974).

Maple seedlings were extremely abundant, occurring at a density of about five per m². while beech seedlings were much less common, with a density of about 0.3 per m² in the 300 transect herb quadrats.

Results of rank-sum tests performed on the indices described in the previous section over individual quad-

Tab. 3. Canopy size distribution by species and quadrat. “Other” species include *Ostrya virginiana*, *Acer rubrum*, and *Liriodendron tulipifera*.

Quadrat	Species	Diam. class (cm)											
		12–15	15–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
1	<i>Acer</i>	7	5	6				1					
	<i>Fagus</i>		1			1	2		1	2			
2	<i>Acer</i>	3	5	2	1	1	1		1				
	<i>Fagus</i>				1		2		1	3	1		
	Other	1											
3	<i>Acer</i>		1					2	2				
	<i>Fagus</i>		1	2	1		1	3			1	1	
4	<i>Acer</i>	1		6				2					
	<i>Fagus</i>	1			1	1	1	2	1				1
5	<i>Acer</i>	1	8	1									
	<i>Fagus</i>	1	1	1			2	3	1	1			
6	<i>Acer</i>		1	1			4	2	1				
	<i>Fagus</i>		2	2	1	1	1	1					
	Other	1	1										
Total	<i>Acer</i>	12	20	16	1	1	5	7	3	1			
	<i>Fagus</i>	2	5	5	4	3	9	9	4	5	3	1	1
	Other	2	1										
Total	All spp.	16	26	21	5	4	14	16	7	6	3	1	1
	Average ha ⁻¹	27	43	35	8	7	23	27	12	10	5	2	2

rats and the total sample are listed in the form of approximate standard normal deviates in Tab. 4. For deviates greater than one, + or – indicates that saplings of one or both species are found at comparatively high levels of influence of canopy of the same species, or of

the other species respectively. Thus, assuming that these saplings are probable replacements for canopy trees, the – suggests a tendency toward reciprocal replacement. These tests were performed for all saplings, then for all saplings except beech root sprouts, the latter results

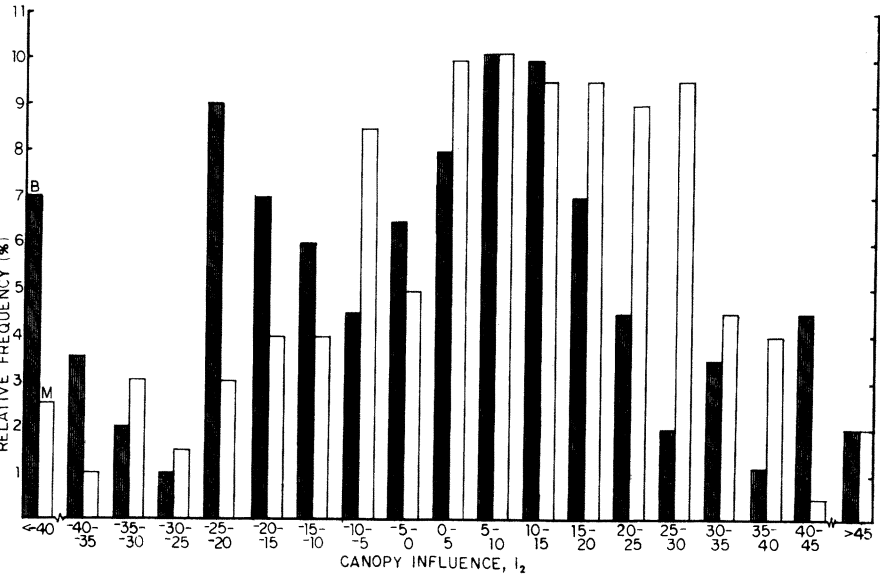


Fig. 1. Distribution of beech and maple saplings with respect to canopy influence as quantified by I_2 . Relative frequency is that proportion of saplings of a species within a given range of the index and sums to 100 for either species. Shaded bars represent beech, unshaded maple.

Tab. 4. Results of rank-sum tests on I_1 and I_2 in form of standard normal deviates. "Modified" results are those for test with rootsprouts removed. Superscript "+" or "-" signify, respectively, conspecific or antispecific association of beech and maple saplings and canopy individuals. Dots indicate significance at the following levels; one dot indicates $\alpha < 0.05$, two dots $\alpha < 0.01$, and three dots $\alpha < 0.005$.

Quadrat	Total	1 Modified	Total	2 Modified
1	0.81	0.51	2.12 ⁺ ●	0.86
2	3.05 ⁻ ●●●	3.14 ⁻ ●●●	2.64 ⁻ ●●	3.14 ⁻ ●●●
3	1.60 ⁺	0.14	2.80 ⁺ ●●	0.56
4	3.63 ⁺ ●●●	2.43 ⁺ ●	2.46 ⁺ ●	1.17 ⁺
5	0.03	0.05	0.60	0.42
6	1.61 ⁻	1.59 ⁻	0.93	1.41 ⁻
Total	0.98	2.27 ⁻ ●	0.26	3.53 ⁻ ●●●

listed as "modified" in Tab. 4. Significance indicates rejection of the H_0 previously stated. H_0 is rejected, by both indices, in favor of reciprocal replacement when all saplings, exclusive of beech root sprouts, are included in the test. The distribution of saplings with respect to I_2 is graphically displayed in Fig. 1.

No understory species were found to be significantly associated with either species of sapling.

It is interesting to note, in Tab. 3, a gap in canopy size distribution from 30 to 50 cm dbh. Cain (1935) observed a similar gap between 6 and 22 inch dbh. Whether these gaps represent a single pulse and what its cause might be are unclear.

4. Discussion

It should be noted that the results shown in Tab. 4 change greatly with removal of beech root sprouts, vary among quadrats, and, with removal of sprouts, show strongly differential response by the two species as saplings to canopy influence. These observations will be discussed in the following paragraphs.

The significance of the ability of beech to root-sprout is not clear. There is some dispute regarding the ability of such sprouts to develop into canopy trees (Fowells 1965). Further, there is great variation over the range of beech in the comparative importance of root sprouts and seedlings in reproduction (Ward 1961). In Warren Woods, near the western border of the range of beech, most reproduction is by seedlings (Ward 1961). The change in test results with removal of sprouts suggests different strategies of dispersal of reproduction by seedlings and sprouts. Only saplings definitely identifiable as sprouts – thirty individuals – were withdrawn from treatment. There were probably other sprouts no longer identifiable as such, having lost their connection to the parent tree. Had they been removed as well, the change in test results might have been greater. Most

identified sprouts were very close to or actually attached to the base of the parent tree (thus their great influence on the distance-weighted I_2) and it is quite possible that they would be killed or damaged by the fall of that tree. It is possible, also, that diseases infecting a dying parent tree might be conveyed to the sapling via such an intimate connection (Campbell 1938). It may be, then, that these sprouts are less likely successors to the canopy than other saplings (Tubbs 1977).

Among the results of tests on individual quadrats, those for quadrats two, four, and six are worthy of further discussion. Fox (1977) suggests that the phenomenon of reciprocal replacement is most apparent in monospecific patches of canopy. Peterson's index of segregation indicated no significant contagion among maple canopy trees in Warren Woods. The index calculated for beech trees was significantly ($\alpha < 0.05$) greater than one, indicating positive contagion among beech canopy individuals. It was suggested by subjective observation that quadrat two was partially in a patch of pure beech canopy and this quadrat yielded test results most indicative of inverse correlation of canopy and sapling species. Quadrat six included an area of high, though not complete, maple dominance, and, again, results are indicative of reciprocal replacement. Quadrat four showed strong suggestion of the reverse tendency. It was strongly dominated by beech canopy but individual maple trees were scattered through it. It will be noted, though, that this quadrat contained relatively few saplings, an unusually high proportion of which were beech. Many of these were clustered around canopy beech trees and, though connecting roots were not apparent, may well have been of root sprout origin.

The very high significance of the rank-sum test when performed on the indices over all saplings exclusive of root sprouts indicates that at least those saplings originating as seedlings show different distributions of beech and maple with respect to the species of influencing canopy trees. This difference is not the one expected were it due only to passive mechanisms such as seed dispersal: if dispersal were the cause young individuals would most likely be near conspecific parent trees. Similarly, if sapling distribution were being determined by microsite factors unrelated to canopy, it would be expected, unless an unlikely reversal of conditions had taken place, that saplings would find most suitable sites like those found most suitable by previous generations of the species, now canopy individuals, and thus be found near these individuals. Test results indicate the reverse affinity. If seeds of either species were dispersed poorly and no further canopy effects influenced sapling distribution, frequencies of saplings of that species would be higher in the range of high conspecific canopy influence. If both species had efficient dispersal mechanisms and further canopy influence were absent, frequency distributions of the two species of saplings should be nearly uninfluenced by canopy influence. As can be seen in Fig. 1, higher frequencies of

beech and maple saplings are relatively displaced towards higher values of antispecific canopy influence as described by I_2 .

There must be, then, canopy-determined differential mortality at some stage of development between seed dispersal and entrance into the sapling size class. In an attempt to discover the age at which this mortality occurs, seedling distributions were analyzed. When seedlings were tested using chi-squared tests for association with saplings of either species, no significant results were obtained for maple seedlings. Beech seedlings, however, showed a significant ($\alpha < 0.05$) association with maple saplings. Since maple saplings appear to be associated with beech canopy, beech seedlings may show similar association. Similarly, when rank-sum tests were performed on seedlings in sapling-centered plots, no significant results were obtained for maple seedlings, and beech seedlings showed a slight ($0.10 > \alpha > 0.05$) tendency to be found at higher beech canopy influence than maple seedlings. Rankings for plots were determined by values of I_2 for the saplings on which they were centered. These results might be predicted for maple, whose seed is easily dispersed, and beech, whose seed is poorly dispersed (Fowells 1965). It would appear, then, that the mortality in question does not occur previous to or in the seedlings stage. This suggests that the mortality-causing factor is not a litter effect or one active in the humus layers inhabited by the seedling root system. A differential effect of canopy-shading might also be expected to manifest itself in the seedling size class, especially since seedlings were frequent for both species in age classes up to 15 yr.

That no significant associations were found for any understory species with either species of sapling, further indicates that surface characteristics of the substrate of suitable sapling microsites do not differ significantly for the two dominant tree species.

Tab. 5 is a contingency table illustrating species of each dead sapling and species of dominant canopy influence at the site. The significance ($\alpha < 0.005$) of these values is obvious. Dead saplings were found primarily beneath conspecific canopy. Of further interest is the fact that all of these saplings were small – between two and four meters in height. It appears that there is high mortality between the seedling and sapling size classes, extending slightly into the range of the latter, and that this mortality is highly concentrated beneath conspecific canopy.

Another problem of interest is whether the general response of positive antispecific correlation of sapling and canopy is a mutual response of beech and maple saplings and canopy or a result of only species of sapling being very sensitive to canopy influence or the influence of one species of canopy on sapling distribution being particularly great. Any of these interactions could account for the significance of the test results in Tab. 4. To approach this question new indices, indicative of single-species canopy influence, were created. The first, I_3 ,

Tab. 5. Contingency table showing species of dead saplings and primary canopy dominant at sites of dead saplings. For one maple sapling there was no obvious dominance of the canopy by either species.

		Dominant canopy species		
		<i>Acer</i>	<i>Fagus</i>	?
Sapling species	<i>Acer</i>	11	0	1
	<i>Fagus</i>	0	6	0

is the distanceweighted influence of beech, calculated for each sapling by the first term in the formula for I_2 . The second, I_4 , is the corresponding sum for maple canopy. Rank-sum tests were performed yielding, for I_3 , a standard normal deviate of 2.5030 and, for I_4 , a deviate of 3.5724. Both of these are highly significant (at $\alpha < 0.02$ and $\alpha < 0.005$ respectively) and indicate association of both species of sapling with antispecific canopy. It is probably not justifiable to interpret the greater significance of the score for I_4 to be the result of greater influence on sapling distribution by maple canopy: this could simply be the result of undetected beech root sprouts. Sapling distribution with respect to these indices is graphed in Fig. 2. Frequencies of both species are skewed to the left on the graph over maple influence in comparison with that over beech influence. This is a result of the greater dominance of beech in the canopy. It is illustrative of the results of the statistical tests to note that, on the axis of beech influence, frequency of beech saplings is generally greater than that of maple at low values of beech influence, while at higher values, the reverse is the case. The reverse holds for sapling distribution along the axis of maple influence with the switchover from higher maple to higher beech sapling frequency occurring rather quickly. Apparently the differential mortality involves interactions between saplings and canopy trees of both species.

Certain determination of the cause of this differential mortality must await experimental data. However, some speculation is not unreasonable. One cause which might be proposed is an allelochemical interaction. Tubbs (1973, 1976) has found allelochemical suppression of several species of seedlings by root exudates of maple seedlings, but has not tested for such suppression of beech or maple seedlings. Such effects, though, would presumably be apparent in seedling size classes and so are probably not the cause of the phenomenon in question. The same argument applies to mechanical litter effects as a cause: such factors would have their major effect on seedlings in early stages of establishment. Dispersal of falling leaves is probably great enough to prevent concentration of litter of one canopy species in such a mixed forest anyway (Stone 1977). The limitation of conspecific understory suggested by Janzen's

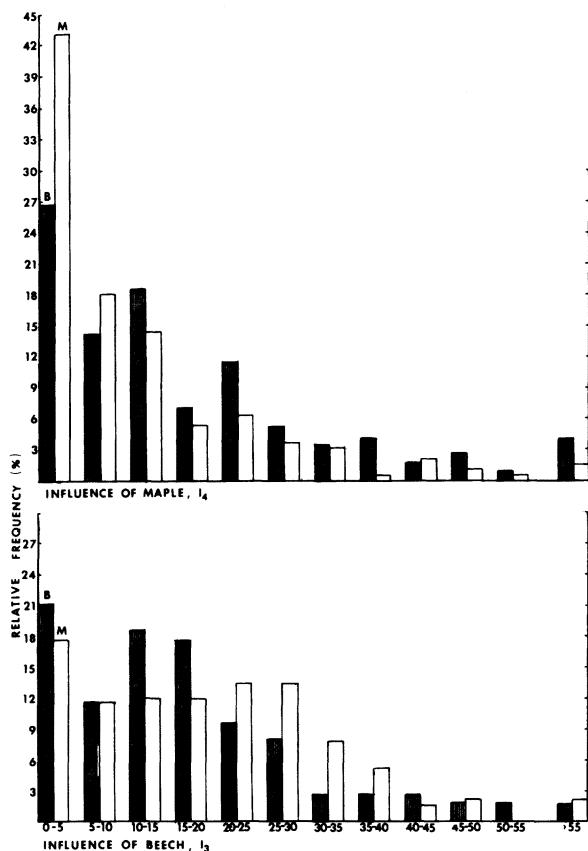


Fig. 2. Distribution of beech and maple saplings with respect to canopy influence of beech and maple as quantified by I_3 and I_4 . Relative frequency as defined for Fig. 1. Shaded bars represent beech saplings, unshaded bars maple saplings.

(1971) predation hypothesis or by influence of disease vectors harbored by conspecific canopy (Fox 1977) would be manifest prior to seedling establishment or in the seedling class. Horn (1971) suggests a mechanism relying on differences in shade tolerance and canopy density, proposing that beech canopy imposes deeper shade than maple while young maples are more shade-tolerant than beeches. This ranking of shade tolerance is not universally agreed upon (Fowells 1965). It is not clear what age group would respond to this mechanism. Again, one would expect some response in the seedling class.

Different species may selectively deplete various soil nutrients, thus altering soil characteristics of subjacent microsites and perhaps affecting survival of seedlings or saplings (Zinke 1962).

It is not unlikely that competition should be especially strong between saplings and conspecific canopy trees. Further, root competition, would be expected to increase greatly in intensity with passage into the sapling stage and concurrent extension of roots from litter layers to deeper soil horizons exploited by mature trees. Canopy individuals would have tremendous advantages

in such competition and might even parasitize saplings via root grafts or mycorrhizal connections. This hypothesis conveniently accounts for the apparent occurrence of differential mortality in or just previous to the small sapling stage.

5. Conclusions

Reciprocal replacement phenomena have been reported for eastern North American beech-maple forests by Horn (1971), Smith (pers. comm.), and Fox (1977). Smith, in his proposal, plotted each sapling in his study on a graph whose abscissa was the distance from that sapling to the nearest canopy individual of maple and ordinate the distance to the nearest canopy individual of beech. He discovered that maple saplings were predominantly below and beech saplings above a diagonal line bisecting the graph through the origin – that is, saplings were, in general, closest to canopy trees of the other species. Though this treatment suggests relationships not clearly apparent upon observation, it has the shortcomings, in this form, of dealing with canopy trees equally regardless of size, second, of similarly disregarding distance from sapling to canopy tree, and, third, of taking into account only two canopy trees for each sapling. If the differential distribution of saplings is effected through modification of sapling microsites by canopy trees, the degree to which a microsite is altered would be dependent on size, distance, and number of influencing canopy individuals. In Fox's (1977) study saplings were judged to be under influence of one or the other canopy species with no consideration of intermediate possibilities or quantification of influence. Much of the study relied on data from monospecific patches and gaps. These shortcomings are partially answered by the indices used in this study.

Two important assumptions are made in this study in accepting the hypothesis of reciprocal replacement for this stand; first, that replacement in Warren Woods is primarily by advance reproduction in the form of suppressed saplings, and, second, that the indices used are representative of canopy influence or at least a monotonic function thereof. The first assumption is supported by the near total absence of intolerant species in the canopy and by the findings of Brewer and Merritt (1978). The second assumption is intuitively reasonable and is supported by the agreement of several indices of different derivation.

A necessary consideration in interpreting the significance of the observed distribution of seedling-derived saplings is of the role of beech root sprouts in canopy replacement. If root sprouts immediately adjacent to parent trees are frequent successors of these trees, the reciprocal replacement interaction could be blocked. The sprouts removed from consideration, however, were few relative to the total number of saplings. They were mostly very near the parent tree, often directly connected to large roots, and might, therefore, have a

high probability of being destroyed when the parent tree is uprooted or infected by pathogens attacking a damaged or dying parent tree. Many large canopy beeches had no apparent root sprouts. Considering these factors and observation of gaps in the field, it is probable that, in this stand, such root sprouts are of minor importance in canopy replacement.

It is, at any rate, clear that a differential mortality rate specific to canopy and sapling species is affecting seedling-derived saplings and leading to an anti-specific association of canopy and understory individuals. The estimate of the significance of this differential in seedling-derived sapling distribution is, in fact, conservative as some unidentified root sprouts were almost certainly included in analyses. Thus, in light of arguments in the last two paragraphs, it appears that in Warren Woods reciprocal replacement is a feasible mechanism for maintenance of canopy codominance.

The generalization of this conclusion, even to other areas of beech-maple forest, must not be made without supportive data. Fox's (1977) proposal of such a phenomenon in beech-maple forests is based also on data from Warren Woods. Smith's data (pers. comm.) is from a similar stand also in southern Michigan. Ward (1956, 1961) has shown that relative importance of seedlings and root-sprouts in beech reproduction varies over the range of the species and with humus type. Furthermore, changes in composition over the range of beech-maple and northern hardwood forest types will influence the interactions of these two species. Forcier (1975), in New Hampshire, found beech saplings associated with maple canopy, but did not detect the reverse association and so suggested a successional trend toward beech dominance. Other data interpretable in this respect from more easterly regions of the joint range of beech and maple are rare.

It is becoming increasingly apparent that understanding of the structure and composition of communities requires study of interactions of the constituent species at all stages of their life cycle. This study elucidates one possible such interaction for one community.

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